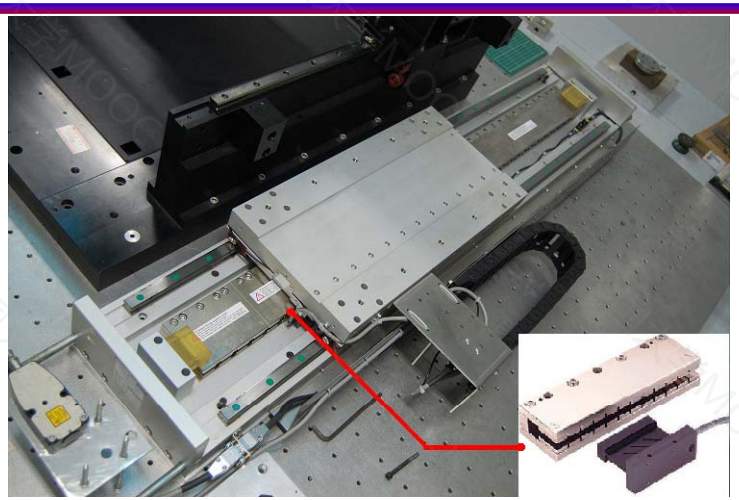
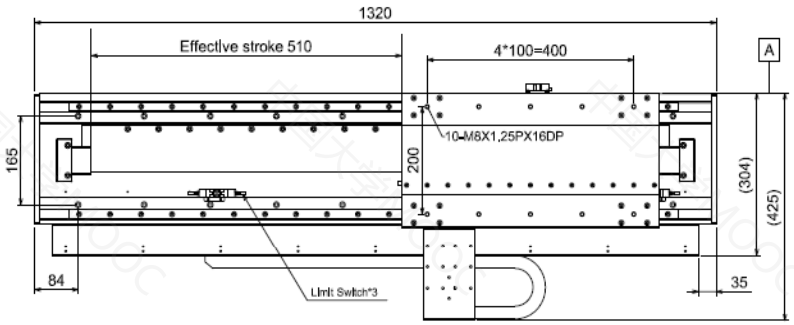
A Case Study:

System Identification of a HIWIN Linear Motor

(案例分析: 直线电机的系统辨识)

Linear Motor Physical Parameters

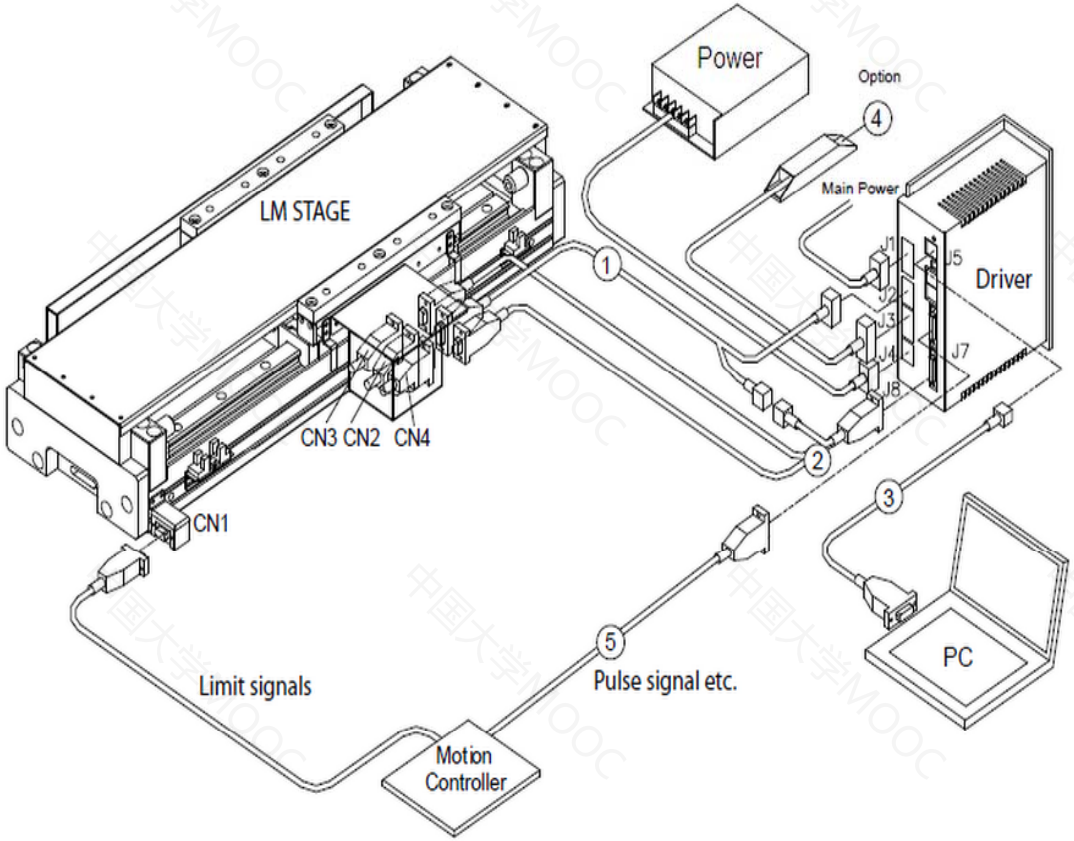




⇨ + Direction

Item	Symbol	Unit	LMCD8
Continuous force	F_c	N	262
Continuous current	I_c	A(rms)	3.25
Peak force(1s)	F_p	N	786
Peak current(1s)	I_p	A(rms)	9.75
Force constant	K_f	N/A	80.6
Max. coil temperature	T_{max}	℃	100
Pole pitch	2τ	mm	60
Bend radius of motor cable	R_{bend}	mm	37.5
Back emf constant	K_v	Vrms/(m/s)	33
Motor constant (25℃)	K_m	$N/\sqrt{W}$	20
Thermal resistance	R_{th}	℃/W	0.21
Mass of forcer	M_f	kg	1.76
Unit mass of stator	M_s	kg/m	16
Electrical time constant	K_e	ms	0.41
Resistance(25℃) (per phase)	R_{25}	Ω	10.8
Inductance (per phase)	L	mH	4.5
Length of forcer	L_f	mm	500/13
Height of forcer	h	mm	87.5
Height of stator	H_s	mm	86.8
Width of stator	W_s	mm	35.5

Control System Setup



Position Measurement System

Linear Encoder (线性编码器): **Mercury II Model SP 4800**

Linear - 20μm grating pitch

Maximum Interpolation	Resolution	Maximum Speed*	Model 4800
x4	5.000μm/count	5000mm/sec	
x10	2.000μm/count	5000mm/sec	
x20	1.000μm/count	5000mm/sec	
x40	0.500μm/count	5000mm/sec	
x80	0.250μm/count	5000mm/sec	
x100	0.200μm/count	5000mm/sec	
x200	0.100μm/count	4000mm/sec	
x400	0.050μm/count	2000mm/sec	
x1000	20.0nm/count	800mm/sec	
x2000	10.0nm/count	400mm/sec	
x4000	5.00nm/count	200mm/sec	
x8000	2.50nm/count	100mm/sec	
x16384	1.22nm/count†	48.8mm/sec	

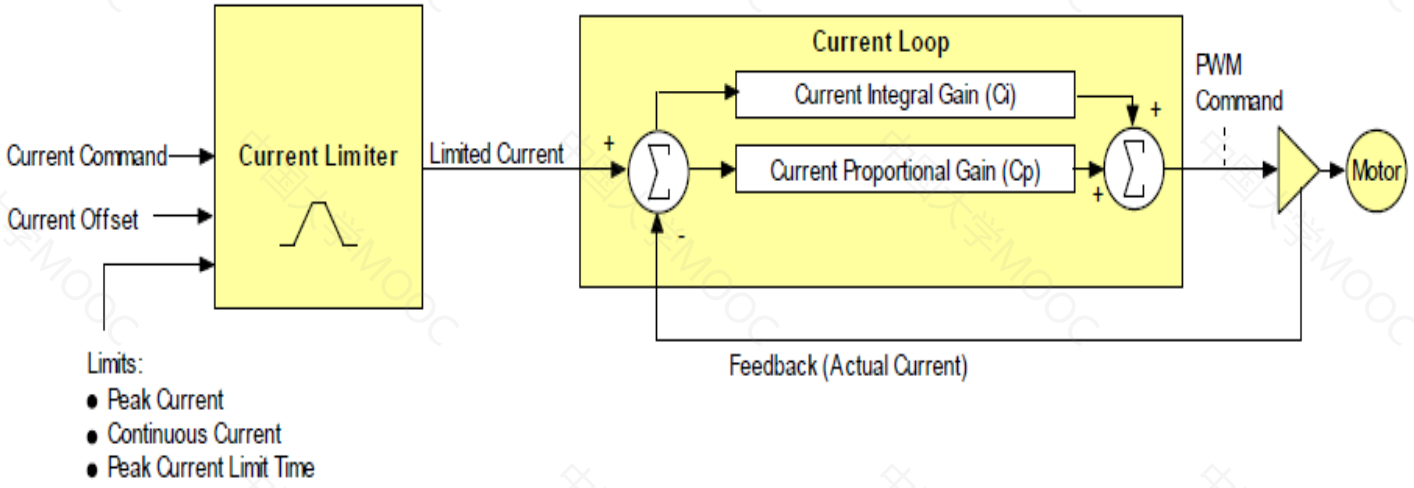
Short-travel accuracy (短行程精度):

± 25nm over any 20μm movement

Long-travel accuracy (长行程精度):

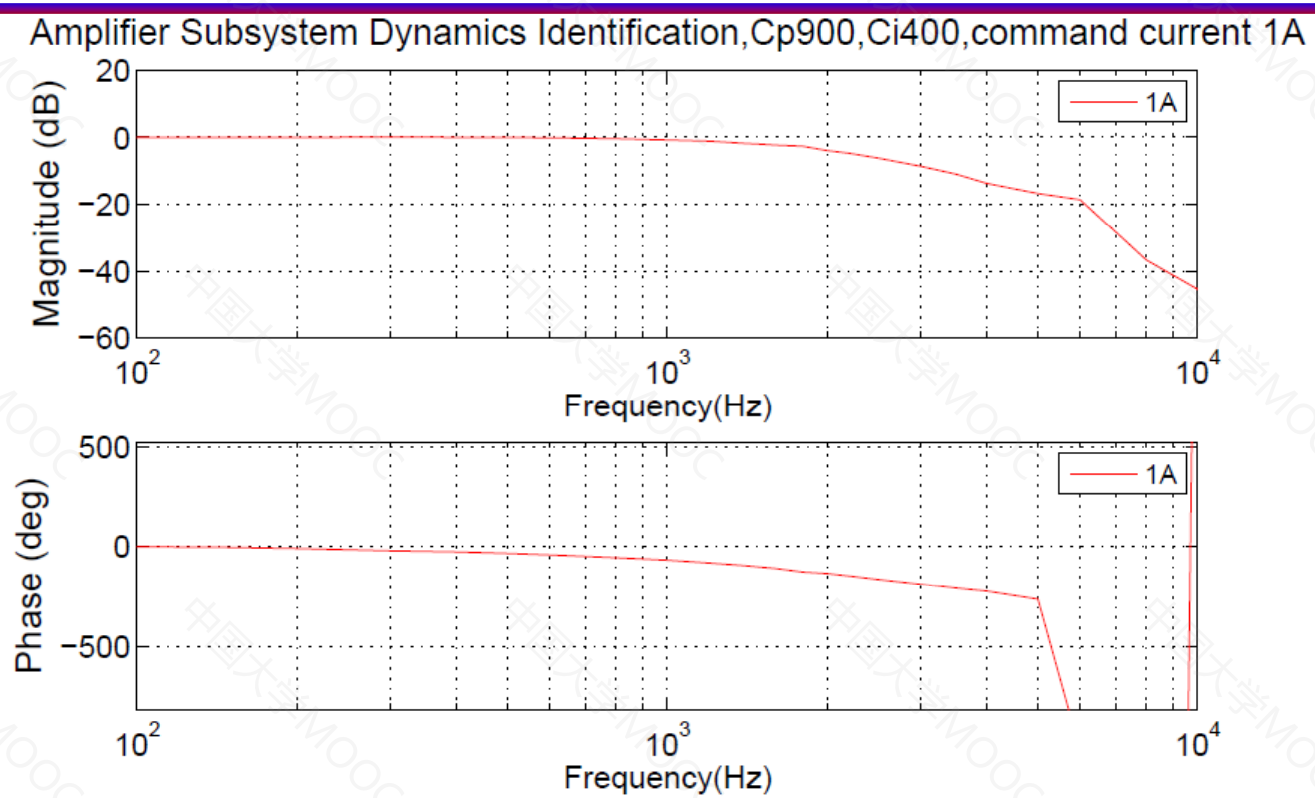
better than ± 2.5μm for scales 155mm to 1m

Motor Amplifier Subsystem



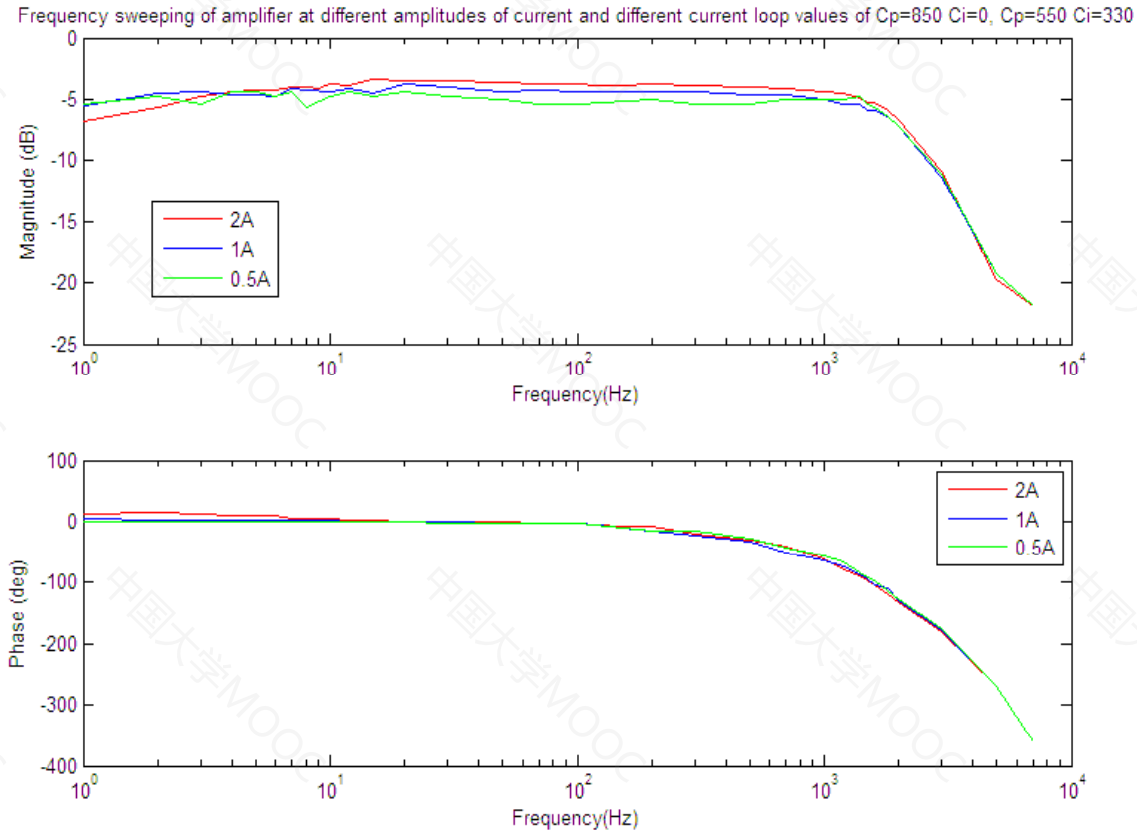
Current loop type (电流环)	100% digital
Sampling rate(time) (采样频率/时间)	15kHz(67μs)

Amplifier Subsystem Dynamics Identification



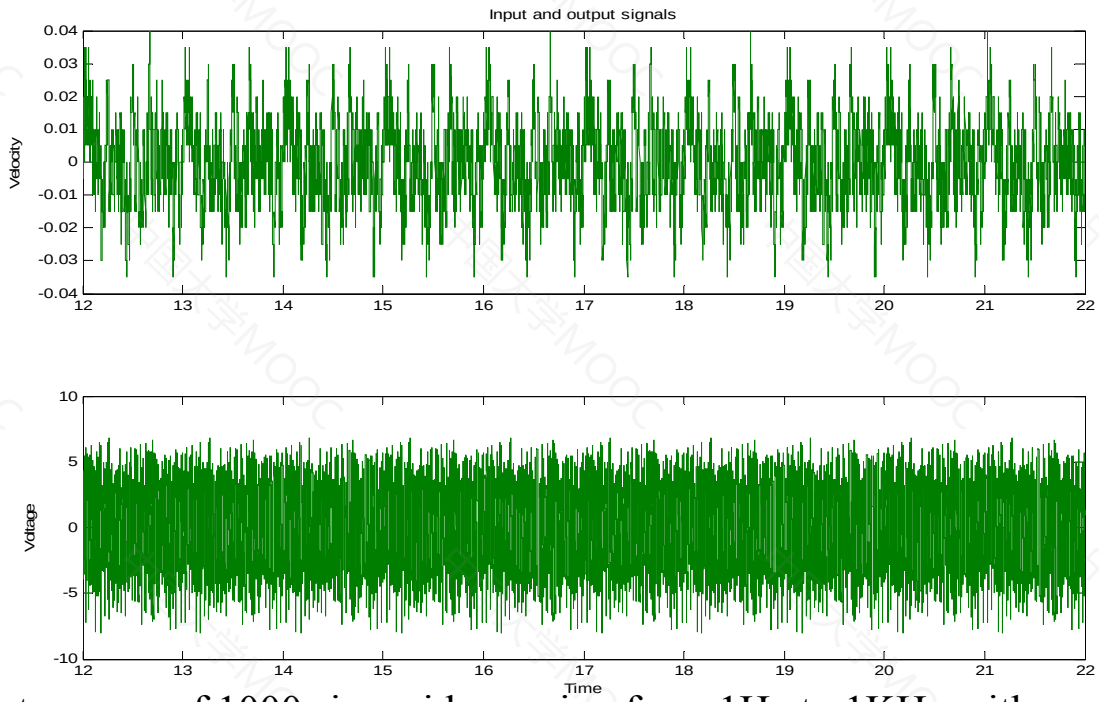
Frequency Response via Sweeping Sinusoidal Tests

Amplifier Subsystem Dynamics Identification



System Dynamics Identification

- **Single Sinusoidal Test**
  (单次正弦测试)



"***idinput***" command to generate a sum of 1000 sinusoids ranging from 1Hz to 1KHz with a sampling frequency of 10KHz **(产生1000个正弦叠加信号，频率从1Hz到1kHz，采样频率10KHz)** :

sinesum = idinput([10000 1 1],'sine', [0 0.2],[-1 1],[1000 50 1])

System Dynamics Identification

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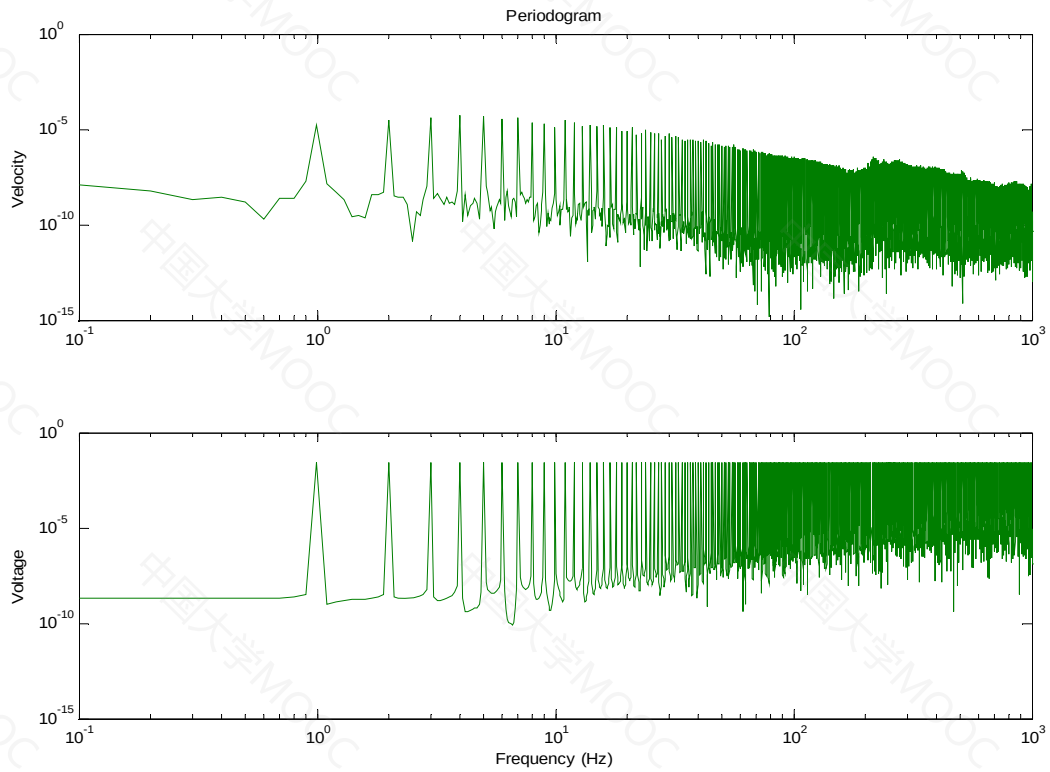
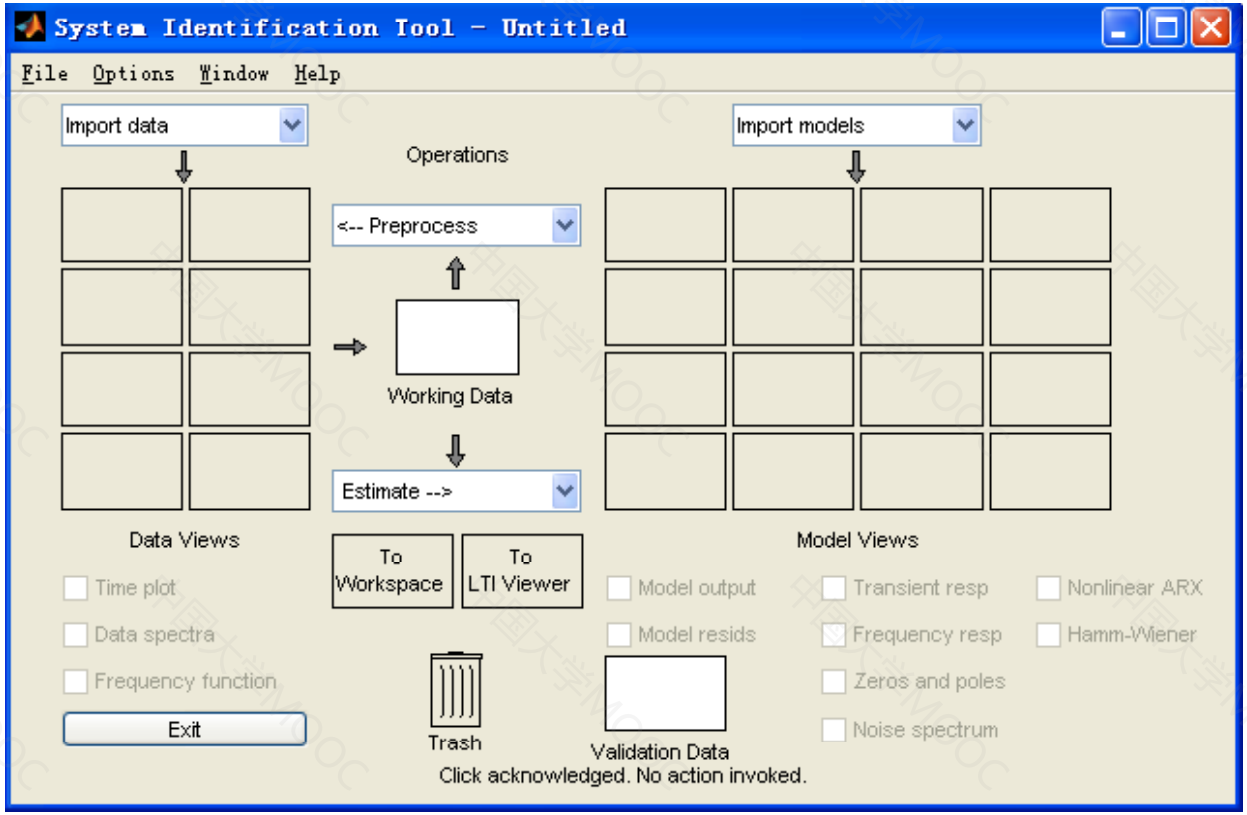


Spectrum of Input/Output using MATLAB "*fft*"

System Identification Tool Box GUI



ident

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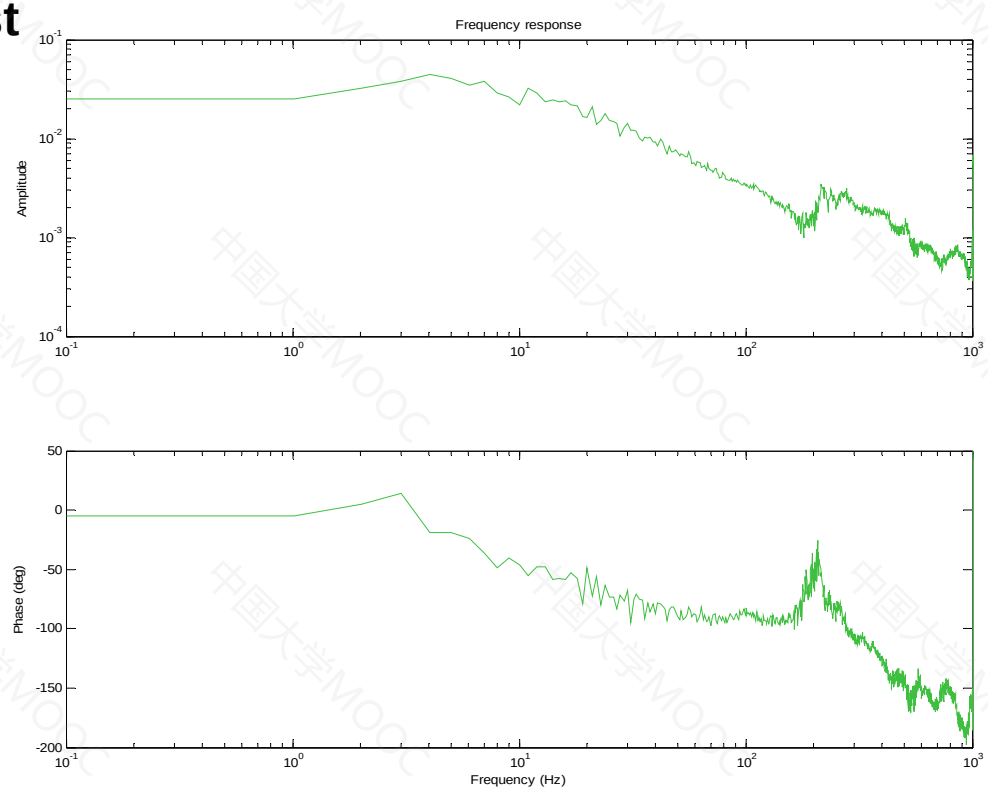

Frequency Response Estimate using MATLAB "***SPAFDR***" with 5000f

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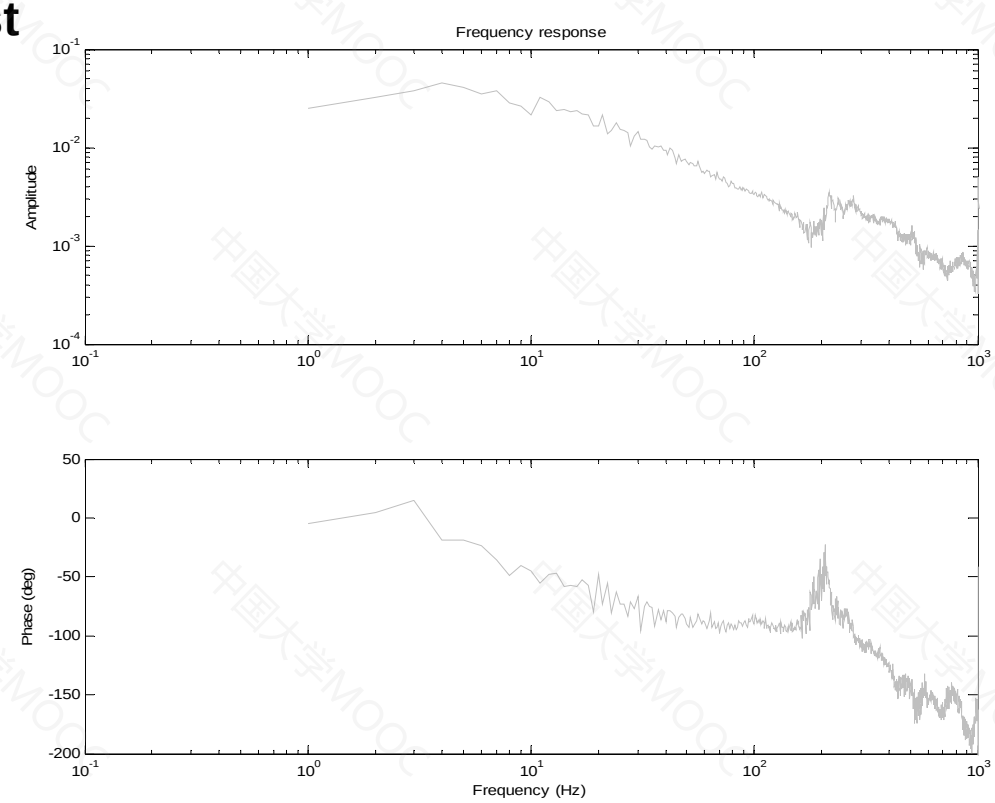

Frequency Response Estimate using MATLAB "***ETFE***" with 5000f

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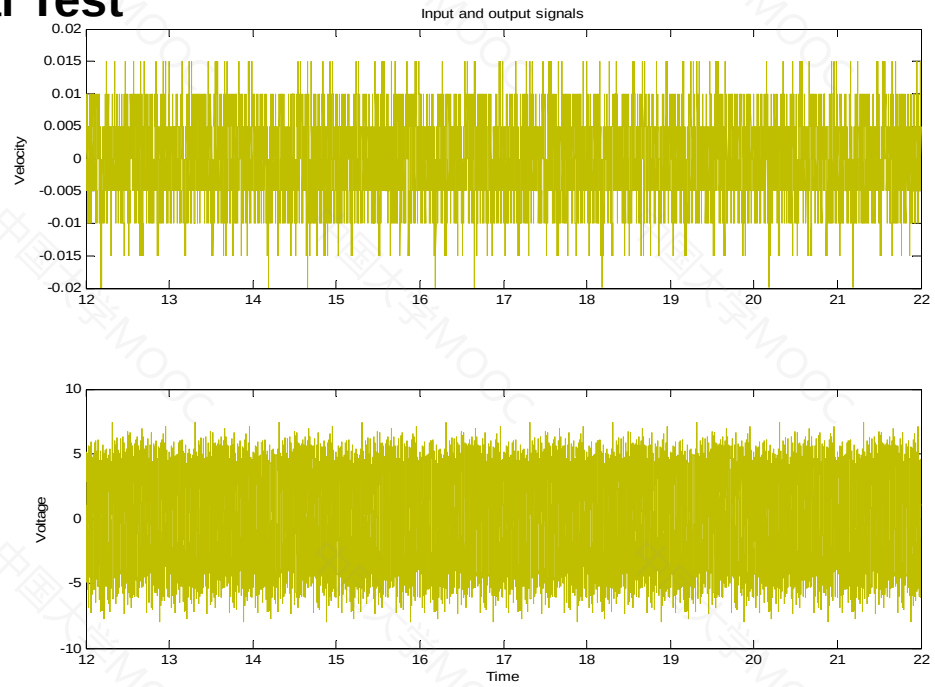

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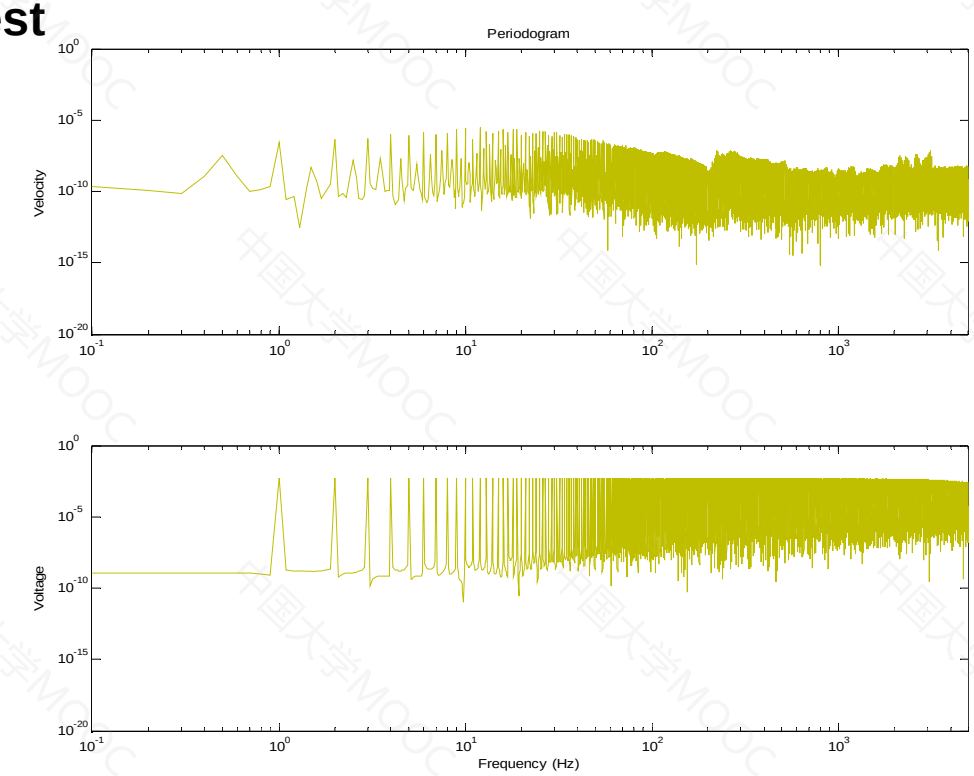

Spectrum of Input/Output using MATLAB "*fft*"

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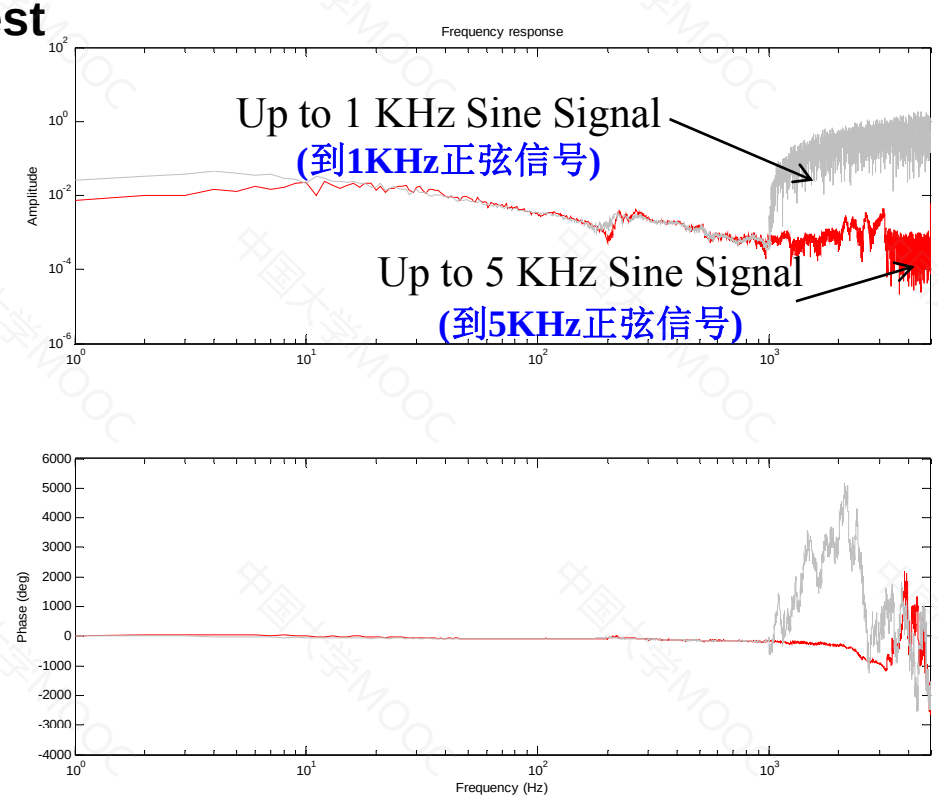

Frequency Response Estimate using MATLAB "***ETFE***" with 5000f

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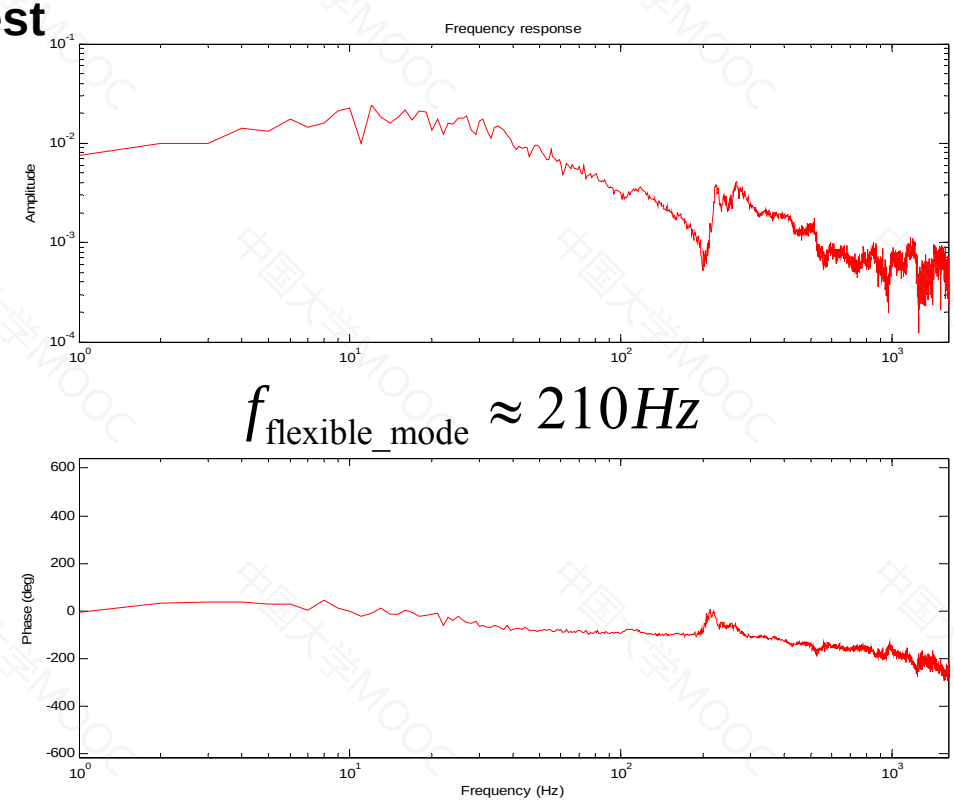


$$f_{\text{flexible_mode}} \approx 210Hz$$

Frequency Response Estimate using MATLAB "***ETFE***" with 5000f

System Dynamics Identification Results

The previous system identification results on Frequency Responses shows that **within 100Hz the order of the system dynamics from input voltage to output position is 2**. Thus, within this frequency range, the system may be modeled adequately by the rigid body dynamics of the stage given by **(之前通过频域响应的系统辨识结果表明系统在100Hz以内从输入电压到输出位置的动态系统是2阶的。 因此，在这个频率以内，可以用刚性动力学来对滑台系统建模)**

$$\dot{x}_1 = x_2$$
$$M\dot{x}_2 = u - Bx_2 - F_{fn}(x_2) - F_r(x_1) + F_d(x_1, x_2, t)$$
$$y = x_1$$

where

x_1: position **(位置)**

x_2: velocity **(速度)**

y: output **(输出量)**

F_{fn}: nonlinear friction **(非线性摩擦力)**

F_d: lumped disturbance **(集中干扰)**

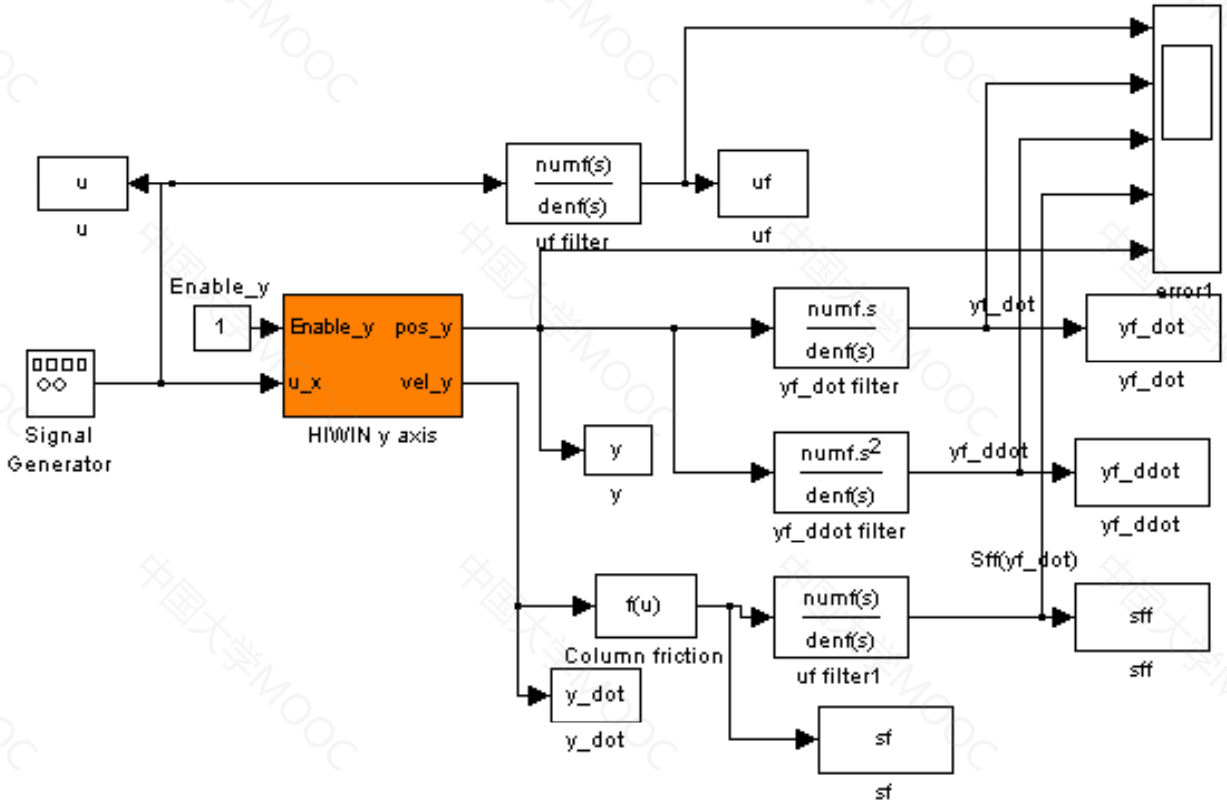
M: mass of load/motor **(负载/电机的质量)**

u: input voltage **(输入电压)**

B: viscous friction const **(粘性摩擦常数)**

F_r: force ripple **(波动力)**

Off-line Parameter Estimations



Off-line Parameter Estimations

OFF-LINE PARAMETER ESTIMATION RESULTS

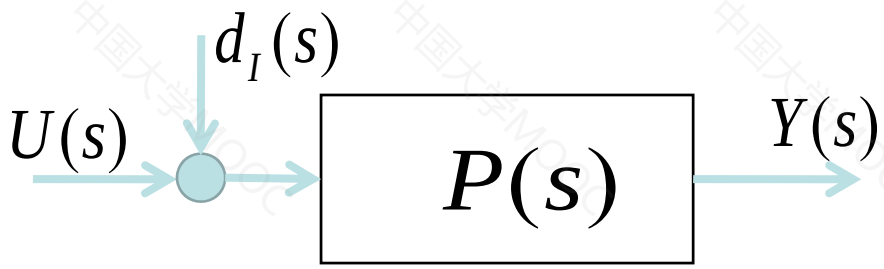
	M	B	A_f	b_n	Δ
right end	0.3001	0.8130	0.2614	0.0065	0.0284
right	0.3004	0.8067	0.2615	0.0057	0.0284
middle	0.2998	0.7900	0.2730	0.0007	0.0319
left	0.3003	0.7858	0.2742	-0.0002	0.0302
left end	0.3004	0.7816	0.2742	0.0001	0.0309
average	0.3002	0.7954	0.2689	0.0026	
right end load	0.3897	0.8395	0.2555	0.0001	0.0322
right load	0.3896	0.8434	0.2557	0.0001	0.0321
middle load	0.3883	0.8455	0.2586	-0.0002	0.0313
left load	0.3904	0.8698	0.2492	-0.0001	0.0222
left end load	0.3891	0.9186	0.2469	-0.0001	0.0245
average load	0.3894	0.8634	0.2532	-0.0001	

- To evaluate the estimated parameters, the input estimation error is defined by RMS (Root Mean Square) as (为评估被估计的参数效果，定义估计误差的均方根为输入估计误差) $\Delta = \sqrt{\frac{\sum (u_f - \phi_f^t \hat{\theta})^2}{N}}$
- The results of △ shown in above table suggest that the model estimation errors for different positions are small and consistent (上表表明不同位置的模型估计误差是较小的且一致的).

Identified Nominal Motor Model (noload)

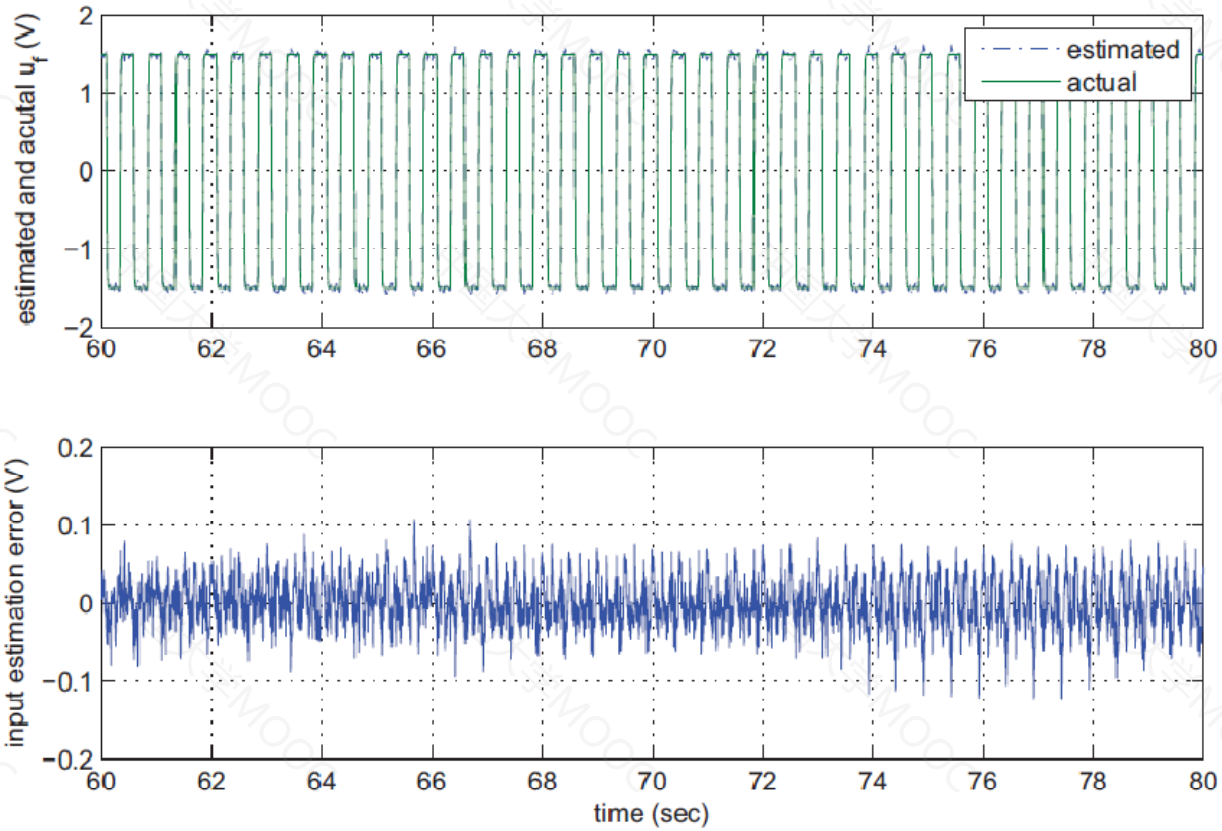
$$0.3\ddot{y} = u - 0.7954\dot{y} - 0.269\,\mathrm{sign}(\dot{y}) - F_r(y) + F_d$$

LTI Model for linear controller designs:



$$K_m = 1.257$$
$$\tau_m = 0.377$$
$$K_b = 3.33$$
$$\omega_m = 2.65$$

$$P(s) = \frac{1}{0.3s^2 + 0.7954s} = \frac{K_m}{s(\tau_m s + 1)} = \frac{K_b}{s(s + \omega_m)}$$

$$d_I = -0.269\,\mathrm{sign}(\dot{y}) - F_r(y) + F_d$$

Model Validation (no load)